체점 과제.

I.

1. $\sum_{n=1}^{\infty} \left(-\frac{1}{n}\right)^n$

$= \sum_{n=1}^{\infty} (-1)^n \cdot \left(\frac{1}{n}\right)^n$

by 1) $a_n = \left(\frac{1}{n}\right)^n$: 감소

교대급수판정법 2) $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(\frac{1}{n}\right)^n = 0$

$\therefore$ 급수는 수렴한다.

or

by 비판정법

$a_n = \left(-\frac{1}{n}\right)^n$

$a_{n+1} = \left(-\frac{1}{n+1}\right)^{n+1}$

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$

$= \lim_{n\to\infty} \left| \frac{\left(-\frac{1}{n+1}\right)\cdot\left(-\frac{1}{n+1}\right)^n}{\left(-\frac{1}{n}\right)^n} \right| = 0 < 1$

2. $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n\sqrt{\ln n}}$

$a_n = \frac{1}{n\sqrt{\ln n}}$

by 교·급·판

a_n : 감소

$\lim_{n\to\infty} a_n = 0$

$\therefore$ 급수는 수렴한다.

3. $\sum_{n=0}^{\infty} \frac{n!}{2\cdot 5\cdot 8 \cdots (3n+2)}$

by 비판정법

$a_n = \frac{n!}{2\cdot 5\cdot 8 \cdots (3n+2)}$, $a_{n+1} = \frac{(n+1)!}{5\cdot 8 \cdots (3n+5)}$

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left| \frac{(n+1)\cdot 2}{3n+5} \right| = \frac{2}{3} < 1$

$\therefore$ 급수는 수렴한다.

4. $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{\ln n}{\sqrt{n}}$

by 교·급·판

$a_n = \frac{\ln n}{\sqrt{n}}$

$\ln n < \sqrt{n}$ ⇒ 분모가 더 크므로 감소

$\lim_{n\to\infty} a_n = 0$

$\therefore$ 급수는 수렴한다.

5. $\sum_{k=1}^{\infty} \frac{k \cdot \ln k}{(k+1)^3}$

$a_k = \frac{k \cdot \ln k}{(k+1)^3}$, $b_k = \frac{k \cdot \ln k}{k^3}$

by 극·비·판

$\lim_{n \to \infty} \frac{a_n}{b_n} = \lim_{n \to \infty} \frac{k^3}{(k+1)^3} = 1 > 0$

$\sum_{k=1}^{\infty} b_k$ 가 수렴하므로

따라서 $\sum_{k=1}^{\infty} \frac{k \cdot \ln k}{(k+1)^3}$ 도 수렴한다.

6. $\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n\sqrt{n}}$

$\tan k = n \Leftrightarrow \tan^{-1} n = k$

$q\pi + \frac{\pi}{4} \le k < \frac{\pi}{2} + q\pi$ (q는 정수)

$\Rightarrow \frac{q\pi + \frac{\pi}{4}}{n\sqrt{n}} \le \frac{\tan^{-1} n}{n\sqrt{n}} < \frac{\frac{\pi}{2} + q\pi}{n\sqrt{n}}$

$\cdots \le \sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n\sqrt{n}} < \sum_{n=1}^{\infty} \frac{\frac{\pi}{2} + q\pi}{n\sqrt{n}}$

↳ p급수에 의해

$p = \frac{3}{2} > 1$ ∴ 수렴.

∴ 위의 급수도 수렴.

7. $\sum_{n=1}^{\infty} \frac{5^n}{3^n + 4^n}$

by 발산 정리

$\lim_{n \to \infty} \frac{5^n}{3^n + 4^n} = \infty \neq 0$

∴ 급수는 발산한다.

II

1. $\sum_{n=1}^{\infty} \frac{(-2)^n \cdot x^n}{\sqrt[4]{n}}$

by 비판정법

$a_n = \frac{(-2)^n \cdot x^n}{\sqrt[4]{n}}$, $a_{n+1} = \frac{(-2)^{n+1} \cdot x^{n+1}}{\sqrt[4]{n+1}}$

$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{\sqrt[4]{n} \cdot x \cdot (-2)}{\sqrt[4]{n+1}} \right| = 2|x| < 1$

$-\frac{1}{2} < x < \frac{1}{2}$

i) $x = -\frac{1}{2}$ … $a_n = \frac{1}{\sqrt[4]{n}}$ … p급수에 의해 $p = \frac{1}{4} < 1$ ∴ 발산

ii) $x = \frac{1}{2}$ … $a_n = \frac{(-1)^n}{\sqrt[4]{n}}$ … ① $b_n = \frac{1}{\sqrt[4]{n}}$: 감소 > by 교·급·판

② $\lim_{n\to\infty} b_n = 0$ ∴ 수렴.

∴ $R = \frac{1}{2}$, $-\frac{1}{2} < x \leq \frac{1}{2}$ 에서 수렴.

2. $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{(x+2)^n}{n \cdot 2^n}$

by 비판정법

$a_n = (-1)^n \cdot \frac{(x+2)^n}{n \cdot 2^n}$, $a_{n+1} = (-1)^{n+1} \cdot \frac{(x+2)^{n+1}}{(n+1) \cdot 2^{n+1}}$

$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{n(x+2)}{(n+1) \cdot 2} \right| = \frac{1}{2}|x+2| < 1$

$|x+2| < 2$

$-4 < x < 0$

i) $x = -4$ … $a_n = \frac{1}{n}$ … p급수에 의해 $p = 1 \leq 1$ ∴ 발산

ii) $x = 0$ … $a_n = \frac{(-1)^n}{n}$ … by 교·급·판 < ① $b_n = \frac{1}{n}$: 감소

② $\lim_{n\to\infty} b_n = 0$

∴ 수렴.

∴ $R = 2$, $-4 < x \leq 0$ 에서 수렴.

3. $\sum_{n=1}^{\infty} n!\,(2x-1)^n$

by 비판정법

$a_n = n!\,(2x-1)^n$, $a_{n+1} = (n+1)!\,(2x-1)^{n+1}$

$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(n+1)\cdot(2x-1)}{1} \right| = \infty$

∴ 급수는 발산한다.

4. $\sum_{n=2}^{\infty} \frac{x^n}{(\ln n)^n}$ by 비판정법

$a_n = \frac{x^n}{(\ln n)^n}$, $a_{n+1} = \frac{x^{n+1}}{(\ln (n+1))^{n+1}}$

$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{x\cdot(\ln n)^n}{(\ln (n+1))^{n+1}} \right| = 0 < 1$

∴ $R = \infty$, $-\infty < x < \infty$ 에서 수렴.

Ⅲ.

1. (오차) $= |S - S_n| \le 0.01$

$\sum_{n=1}^{\infty} (-\frac{1}{n})^n = \sum_{n=1}^{\infty} (-1)^n \cdot (\frac{1}{n})^n$

$a_n = (\frac{1}{n})^n$

$a_3 = \frac{1}{27}$, $a_4 = \frac{1}{256}$

$\frac{1}{256} < \frac{1}{100}$

∴ $n+1 = 4$

$n = 3$ //

2. (오차) $= |S - S_n| < 0.01$

$\sum_{n=1}^{\infty} \frac{(-2)^n}{n!} = \sum_{n=1}^{\infty} \frac{2^n}{n!} (-1)^n$

$a_n = \frac{2^n}{n!}$

$a_6 = \frac{64}{720}$, $a_5 = \frac{32}{120}$ ⇒ $a_5 > 0.01$, $a_6 < 0.01$

∴ $n+1 = 6$

$n = 5$

3. (오차) $= |S - S_n| < 0.0001$

$a_n = \frac{1}{n^5}$

$a_6 > 0.0001$, $a_7 < 0.0001$

$\therefore$ 마지막 $= 7$

$n = 6$

$S_6 = 1 - \frac{1}{32} + \frac{1}{243} - \frac{1}{1024} + \frac{1}{3125} - \frac{1}{7776}$

$\therefore$ 0.9720 //

IV.

1. $f(x) = \frac{x^2}{2+x}$

$= \frac{1}{2\left(1-\left(-\frac{x}{2}\right)\right)} \cdot x^2$

$= \frac{\frac{1}{2}x^2}{\left(1-\left(-\frac{x}{2}\right)\right)}$

$\therefore f(x) = \sum_{n=0}^{\infty} \frac{1}{2}x^2 \cdot \left(-\frac{x}{2}\right)^n$

$n=6 \Rightarrow \frac{1}{2}x^2 \cdot \left(-\frac{x}{2}\right)^6 = \frac{1}{2}x^2 \cdot \frac{x^6}{64}$

$= \frac{1}{128}x^8$

$\therefore \frac{1}{128}$

2. $f(x) = \frac{x^3}{(x-2)^2} = \frac{1}{(x-2)^2} \cdot x^3$

$\left(\frac{-1}{x-2}\right)' = \frac{1}{(x-2)^2}$

$\frac{-1}{x-2} = \frac{-1}{-2\left(1-\frac{x}{2}\right)} = \frac{\frac{1}{2}}{1-\frac{x}{2}} = \sum_{n=0}^{\infty} \frac{1}{2} \cdot \left(\frac{x}{2}\right)^n$

$\Rightarrow f(x) = \sum_{n=0}^{\infty} \frac{1}{2} \cdot n \cdot \left(\frac{x}{2}\right)^{n-1} \cdot \frac{1}{2} \cdot x^3 = \sum_{n=0}^{\infty} \frac{1}{4} n x^3 \left(\frac{x}{2}\right)^{n-1}$

$n=6 \Rightarrow \frac{1}{4} \cdot 6 \cdot x^3 \cdot \left(\frac{x}{2}\right)^5 = \frac{1}{128} \cdot 6 \cdot x^8$ $\therefore \frac{3}{64}$ //